Solution Requirements

Date	13 November 2026
Team ID	0157CS241131
Project Name	AI Travel Itinerary Planner (Trip Genius AI)
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Email/password sign-up, login, session restore and logout through Supabase Auth; the backend verifies the Bearer token on every private call	High
2	Authorization levels	One role: logged-in user. Row Level Security allows a user to view, edit or delete only their own trips and itineraries	High
3	External interfaces	Groq API (openai/gpt-oss-120b) for generation; Supabase REST/Auth for identity and storage; Vercel and Render hosting	High
4	Transactions processing	Generate request then validated prompt then AI JSON then validated response; save creates a trips row plus an itineraries row; update and delete supported	High
5	Reporting	Day-wise itinerary view and downloadable PDF export (jsPDF)	Medium
6	Business rules, etc.	End date ≥ start date; max 30 days; at least one interest; destination required; AI must return exactly the requested number of days with 3+ activities per day	High
7	Compliance to laws or regulations	Passwords are handled by Supabase Auth and not stored by our code. Secrets stay in server environment variables. The AI plan is only a suggestion, not booking advice.	Medium
8	Other	Responsive UI for desktop and mobile; clear error messages for AI timeout, rate limit and invalid output	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	A loading message is shown while the itinerary is generated; other endpoints respond quickly	Non-AI API < 2 s; generation < 30 s (depends on Groq)
2	Scalability	The FastAPI service is stateless, so more instances can be added on Render	Multiple Gunicorn/Uvicorn workers supported
3	Security & Data Privacy	JWT verification, RLS policies, CORS allow-list, no service-role key in the frontend, HTTPS in production	0 secrets in frontend bundle
4	Reliability & Availability	AI timeout, rate limit, empty reply or invalid JSON return a proper HTTP error	An error does not stop the website

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
5	Usability & Accessibility	Simple planner form, interest buttons and a clear colour contrast	Works at 320 px width and above
6	Other	Maintainability: layered backend (routes, services, schemas, db); automated pytest suite (34 tests)	Typed frontend (TypeScript)